Predicting Building - Level Flood Risk by Analysis Comparison of Multi-Layer Perceptron and Bayesian Logistic Regression

DS4420 Project by: Lang Shao, Jiada Li

Northeastern University

Introduction & Problem

•**The Problem:** Current FEMA Flood Insurance Rate Maps (FIRMs) rely on macro-level topographic data, largely ignoring specific physical characteristics of individual structures.

•**The Consequence:** Two houses in the exact same flood zone can have drastically different true vulnerabilities depending on micro-level factors (like foundation height).

•**Our Objective:** Leverage National Structure Inventory (NSI) data to model flood risk at the individual building level, shifting from geospatial area-mapping to structural vulnerability classification.

Data Overview

Dataset: National Structure Inventory (NSI).

FROM GPKG TO CSV!

Target Variable: FEMA FIRM Zones.

•**High-Risk (1):** VE, AE, A, AO, AH.

•**Low-Risk (0):** Area X.

•**Key Features:** Ground elevation, foundation height, structural value, year built, square footage, number of stories, building type, and foundation type.

•Download and documentation:

https://www.hec.usace.army.mil/confluence/nsi/technicalreferences/latest/technical-documentation

References

Demissie, Z., et al. (2024). *Applied Computing and Geosciences, 23*, 100183.

Chen, Y., et al. (2023). *Frontiers in Earth Science, 11*, 1117004.

Ouma, Y. O., & Lawrence, O. (2023). *Int. J. of Intelligent Systems 2023*, 5672401.

Qin, X., et al. (2025). *npj Urban Sustainability, 5*, 19.

U.S. Army Corps of Engineers. *National Structure Inventory (NSI) Technical Documentation.*

Methodology

1. Custom MLP (By Lang)

•**Implementation:** It was built manually by using Python and NumPy matrix operations

Architecture: * Input Layer: d dimensional feature space.

$$h = \max(0, XW_1 + b_1)$$

- Hidden Layer: 16 nodes with **ReLU** activation
- Output Layer: 1 node with **Sigmoid** activation for probability mapping

$$\hat{y} = \sigma(hW_2 + b_2) = \frac{1}{1 + e^{-(hW_2 + b_2)}}$$

•**Optimization:** Full-batch Gradient Descent minimizing Binary Cross-Entropy (BCE) loss (eta=0.05, 2000 epochs) :

$$J = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

2. Bayesian Logistic Regression(By Jiada)

•**Implementation:** built using R and rstanarm library

•**Architecture:** * Input: d dimensional feature space.

- Prior: Normal distribution with mean=0, std= 2.5 for features. Mean=0, std=10 for intercept $w \sim N_p(w^{(0)}, \Sigma_0)$
- Likelihood: Follows a Bernoulli distribution $y_i | w, x_i \sim Bern(\theta = \sigma(w^T x_i))$

•**Optimization:** Use MCMC sampling to retrieve feature weights with iter=1000, chain=4, warmup=500,

Results & Interpretation

1. MLP Results

•**Predictive Metrics:** * Accuracy: **93.97%**

- Precision: 0.9243
- Recall: **0.9440**
- F1-Score: 0.9341

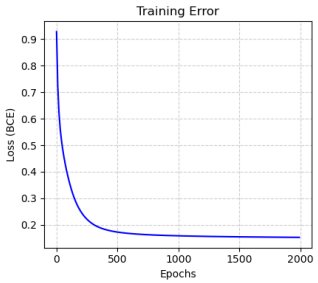
2. Bayesian Logistic Regression Results

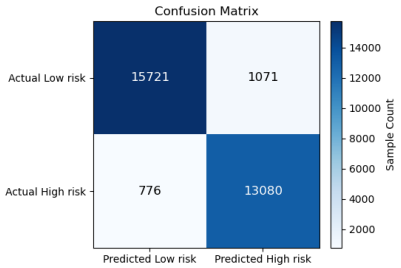
•**Predictive Metrics:** * Accuracy: 92.52%

- Precision: 0.8890
- Recall: 0.7675
- F1-Score: 0.8238

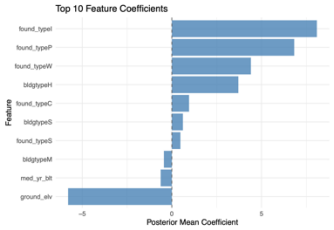
•**Key Takeaway:** The MLP outperforms the Bayesian Logistic Regression across all metrics, achieving higher accuracy (93.97% vs 92.52%), recall (0.9440 vs 0.7675), and F1-score (0.9341 vs 0.8238). The exceptionally high Recall indicates a very low rate of False Negatives. This ensures vulnerable structures are not mistakenly classified as safe, which is paramount for municipal planning.

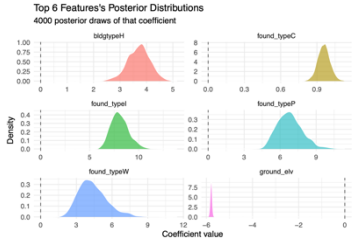
Figures of MLP Results





Figures of BLR Results





Conclusions & Future Work

Model Trade-offs: The MLP demonstrated superior capability in capturing complex, non-linear structural patterns (high accuracy). However, it suffers from a "black-box" nature compared to the probabilistic, easily interpretable coefficients provided by the Bayesian approach. The BLR trades some predictive performance for interpretability and uncertainty quantification. Its posterior coefficients reveal that foundation type and ground elevation are the strongest drivers of flood risk, insights that the MLP cannot provide.

Future Directions: * Expand the dataset to include diverse geographical regions.

•Incorporate a post-hoc interpretability framework, such as **SHAP values**, to understand the marginal contribution of individual building attributes (e.g., ground elevation) to the MLP's predictions. The BLR model has higher interpretability of the features and uncertainty. The combination of a neural network and a Bayesian model (BNN) could improve the predictability.